RSMC-50K

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Closed-Loop Robotic Surgery Motor Controller

Medical Robotics / Precision Motion Control | Rev A2 - Controlled Review Baseline Program Date TBC | Architecture, Atomic Requirements, Interfaces & Verification

Mission profile. Ultra-low-latency motor controller for surgical manipulator joints with sub-millimeter precision intent, FPGA-based field-oriented control and a 50 kHz+ current loop. Motion authority is bounded by hardware-enforced force/torque limits and an independent Safe Torque Off path.

Engineering baseline. The baseline uses deterministic VHDL/C logic, low-voltage DC power, conduction cooling, a hardwired internal backplane and sterilizable EMI-shielded joint modules. Encoder offset, friction and force/torque calibration are configuration-controlled; lifecycle monitoring uses motor-current signatures as an indicator of gearbox wear.

Motor, gearbox, encoder, force sensor, FPGA, power semiconductor, joint enclosure, sterilization process and robot-host interface are controlled variants. Numerical jitter, STO/force response, precision, thermal, acoustic and calibration acceptance limits remain TBC until program-approved evidence closes them. System-level medical-device risk management, regulatory classification, IEC 60601-1/IEC 62304 applicability, usability and sterilization validation are owned by the parent surgical-robot program; this controller PRD provides the subsystem requirements, interfaces and objective evidence needed for that system-level compliance case.

Definition	Controlled product interpretation
Product class	Closed-Loop Robotic Surgery Motor Controller.
Safety / assurance	Ultra-low-latency motor controller for surgical manipulator joints with sub-millimeter precision intent, FPGA-based field-oriented control and a 50 kHz+ current loop.
Documentation baseline	Program Date TBC - Rev A2 controlled review baseline; external release is blocked until the program approves the documentation baseline date.
Primary architecture	The baseline uses deterministic VHDL/C logic, low-voltage DC power, conduction cooling, a hardwired internal backplane and sterilizable EMI-shielded joint modules. Encoder offset, friction and force/torque calibration are configuration-controlled; lifecycle monitoring uses motor-current signatures as an indicator of gearbox wear.
Release philosophy	Motor, gearbox, encoder, force sensor, FPGA, power semiconductor, joint enclosure, sterilization process and robot-host interface are controlled variants. Numerical jitter, STO/force response, precision, thermal, acoustic and calibration acceptance limits remain TBC until program-approved evidence closes them. System-level medical-device risk management, regulatory classification, IEC 60601-1/IEC 62304 applicability, usability and sterilization validation are owned by the parent surgical-robot program; this controller PRD provides the subsystem requirements, interfaces and objective evidence needed for that system-level compliance case.

PRODUCT GOVERNANCE, MOTION AUTHORITY & REQUIREMENT QUALITY

The motor controller shall separate surgical motion control, force/torque limits, STO, calibration, service and non-surgical interfaces with explicit authority precedence.

If a non-safety software path requests motion outside the released authority hierarchy, the system shall deny the motion request and preserve independent hardware force/STO authority.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-001	The motor controller shall separate surgical motion control, force/torque limits, STO, calibration, service and non-surgical interfaces with explicit authority precedence.	architecture review, authority matrix and negative command testing
RSMC50K-002	If a non-safety software path requests motion outside the released authority hierarchy, the system shall deny the motion request and preserve independent hardware force/STO authority.	architecture review, authority matrix and negative command testing
RSMC50K-003	A change to motion interface, safety partition, service tool or controller topology shall reopen architecture review, authority matrix and negative command testing before production release.	architecture review, authority matrix and negative command testing

Verification, margin and future-change rule

Future variants affecting motion interface, safety partition, service tool or controller topology are screened against the mechanism on this page. The release owner may retain prior evidence only where architecture review, authority matrix and negative command testing confirms the similarity boundary.

SYSTEM BOUNDARY, ROBOT JOINT / DRIVE / HOST OWNERSHIP

The controller shall define electrical and timing boundaries among host kinematics, joint sensors, motor, brake, power stage and independent safety functions.

If ownership of torque, position, brake or STO command is ambiguous, the product shall remove motion authority until the interface state is valid and diagnosable.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-004	The controller shall define electrical and timing boundaries among host kinematics, joint sensors, motor, brake, power stage and independent safety functions.	system boundary diagram, ICD and authority/timeout review
RSMC50K-005	If ownership of torque, position, brake or STO command is ambiguous, the system shall remove motion authority until the interface state is valid and diagnosable.	system boundary diagram, ICD and authority/timeout review
RSMC50K-006	Evidence reuse after robot host, motor, encoder, brake, harness or joint architecture is permitted only when system boundary diagram, ICD and authority/timeout review demonstrates that the affected assumptions remain valid.	system boundary diagram, ICD and authority/timeout review

Verification, margin and future-change rule

The change trigger for this page is robot host, motor, encoder, brake, harness or joint architecture. The impact record identifies moved assumptions and the exact system boundary diagram, ICD and authority/timeout review needed for closure.

OPERATING MODES, STARTUP / HOMING / SHUTDOWN

Controlled states shall cover boot, self-test, calibration, homing, enabled motion, limited motion, fault stop, STO and service.

Verification focuses on the point where a prerequisite, homing result or mode transition cannot be validated. The required containment is to prevent torque enable or enter the allocated safe state.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-007	Controlled states shall cover boot, self-test, calibration, homing, enabled motion, limited motion, fault stop, STO and service.	state-machine test, transition fault injection and restart validation
RSMC50K-008	The design shall detect a prerequisite, homing result or mode transition cannot be validated and shall prevent torque enable or enter the allocated safe state.	state-machine test, transition fault injection and restart validation
RSMC50K-009	Startup sequence, homing method, brake logic, calibration flow or service mode shall be treated as a controlled engineering change; approval requires state-machine test, transition fault injection and restart validation.	state-machine test, transition fault injection and restart validation

Verification, margin and future-change rule

Changes involving startup sequence, homing method, brake logic, calibration flow or service mode are engineering events, not editorial updates; approval depends on the targeted state-machine test, transition fault injection and restart validation listed here.

LOW-VOLTAGE DC INPUT, POWER TREE & PROTECTION

The low-voltage DC input shall provide protected rails for power stage, FPGA/processor, sensors and gate drivers across the released robot supply envelope.

If input transient, brownout or rail-sequencing fault threatens controlled gate or logic state, the system shall disable torque generation through the validated power-fault path and preserve fault capture without relying on an unrelated convenience function.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-010	The low-voltage DC input shall provide protected rails for power stage, FPGA/processor, sensors and gate drivers across the released robot supply envelope.	power transient DVP, sequencing test and brownout injection
RSMC50K-011	Loss or corruption involving input transient, brownout or rail-sequencing fault threatens controlled gate or logic state shall not permit silent normal operation; the system shall disable torque generation through the validated power-fault path and preserve fault capture.	power transient DVP, sequencing test and brownout injection
RSMC50K-012	Production equivalence following input voltage, DC/DC converter, PMIC, bulk capacitor or rail sequence requires closure of power transient DVP, sequencing test and brownout injection.	power transient DVP, sequencing test and brownout injection

Verification, margin and future-change rule

For input voltage, DC/DC converter, PMIC, bulk capacitor or rail sequence, the configuration owner documents evidence reuse, new risks and the required power transient DVP, sequencing test and brownout injection before sign-off.

MOTOR INVERTER POWER STAGE & DEVICE MARGIN

The inverter shall deliver the required bidirectional phase current with device voltage/current/thermal margin across the approved motor and motion envelope.

Shoot-through, overcurrent or power-device fault is detected is treated as a diagnosable abnormal condition. The system response is to remove gate authority through the fast hardware path and prevent uncontrolled re-energization.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-013	The inverter shall deliver the required bidirectional phase current with device voltage/current/thermal margin across the approved motor and motion envelope.	power-stage WCCA, double-pulse/fault testing and thermal verification
RSMC50K-014	When shoot-through, overcurrent or power-device fault is detected, the product shall remove gate authority through the fast hardware path and prevent uncontrolled re-energization.	power-stage WCCA, double-pulse/fault testing and thermal verification
RSMC50K-015	The released configuration shall not be extended across power semiconductor, topology, DC bus, motor, switching frequency or layout without power-stage WCCA, double-pulse/fault testing and thermal verification.	power-stage WCCA, double-pulse/fault testing and thermal verification

Verification, margin and future-change rule

Any modification to power semiconductor, topology, DC bus, motor, switching frequency or layout is accepted only after the affected assumptions are rechecked and power-stage WCCA, double-pulse/fault testing and thermal verification is closed.

FPGA / PROCESSOR PARTITION & DETERMINISTIC OWNERSHIP

FPGA and processor responsibilities shall be partitioned so deterministic current/safety functions do not depend on variable-latency noncritical software.

The design prevents silent continuation when processor overload or software stall affects a supervised control exchange; it must instead maintain bounded FPGA safety behavior, revoke invalid motion authority and expose the fault.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-016	FPGA and processor responsibilities shall be partitioned so deterministic current/safety functions do not depend on variable-latency noncritical software.	partition review, overload test and cross-domain fault injection
RSMC50K-017	The released implementation shall maintain bounded FPGA safety behavior, revoke invalid motion authority and expose the fault whenever processor overload or software stall affects a supervised control exchange.	partition review, overload test and cross-domain fault injection
RSMC50K-018	Changes to FPGA device, processor, bus, task allocation or hardware/software boundary shall trigger impact review of affected assumptions and closure of partition review, overload test and cross-domain fault injection.	partition review, overload test and cross-domain fault injection

Verification, margin and future-change rule

The release baseline remains valid across FPGA device, processor, bus, task allocation or hardware/software boundary only within demonstrated similarity. Otherwise, partition review, overload test and cross-domain fault injection is repeated or updated.

PHASE-CURRENT SENSING, ADC & OFFSET DIAGNOSTICS

Phase-current measurement shall meet the accuracy, latency and common-mode requirements needed by current control and hardware force limits.

If current sensor saturation, offset drift or ADC plausibility failure is detected, the system shall limit or disable torque according to the safety allocation and prevent invalid feedback from driving FOC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-019	Phase-current measurement shall meet the accuracy, latency and common-mode requirements needed by current control and hardware force limits.	metrology WCCA, offset/saturation tests and fault injection
RSMC50K-020	If current sensor saturation, offset drift or ADC plausibility failure is detected, the system shall limit or disable torque according to the safety allocation and prevent invalid feedback from driving FOC.	metrology WCCA, offset/saturation tests and fault injection
RSMC50K-021	A change to current sensor, shunt, amplifier, ADC, filter or sampling point shall reopen metrology WCCA, offset/saturation tests and fault injection before production release.	metrology WCCA, offset/saturation tests and fault injection

Verification, margin and future-change rule

Future variants affecting current sensor, shunt, amplifier, ADC, filter or sampling point are screened against the mechanism on this page. The release owner may retain prior evidence only where metrology WCCA, offset/saturation tests and fault injection confirms the similarity boundary.

ENCODER / POSITION SENSOR INTERFACE

Position feedback shall include signal-integrity, plausibility and loss-of-feedback diagnostics appropriate to the released encoder technology.

If encoder data is stale, contradictory, out of range or lost, the product shall prevent unconstrained motion and transition to the allocated limited or safe state.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-022	Position feedback shall include signal-integrity, plausibility and loss-of-feedback diagnostics appropriate to the released encoder technology.	encoder fault injection, plausibility tests and timing analysis
RSMC50K-023	If encoder data is stale, contradictory, out of range or lost, the system shall prevent unconstrained motion and transition to the allocated limited or safe state.	encoder fault injection, plausibility tests and timing analysis
RSMC50K-024	Evidence reuse after encoder type, resolution, interface, cable, interpolation or decoding logic is permitted only when encoder fault injection, plausibility tests and timing analysis demonstrates that the affected assumptions remain valid.	encoder fault injection, plausibility tests and timing analysis

Verification, margin and future-change rule

The change trigger for this page is encoder type, resolution, interface, cable, interpolation or decoding logic. The impact record identifies moved assumptions and the exact encoder fault injection, plausibility tests and timing analysis needed for closure.

FIELD-ORIENTED CONTROL & COMMUTATION

FOC shall use validated motor parameters, current feedback and position/angle information while respecting independent current and force limits.

Verification focuses on the point where angle, parameter or current-vector validity is lost. The required containment is to suppress normal torque production and enter the controlled fault response.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-025	FOC shall use validated motor parameters, current feedback and position/angle information while respecting independent current and force limits.	FOC model correlation, parameter sweep and sensor-fault testing
RSMC50K-026	The design shall detect angle, parameter or current-vector validity is lost and shall suppress normal torque production and enter the controlled fault response.	FOC model correlation, parameter sweep and sensor-fault testing
RSMC50K-027	Motor, FOC algorithm, motor constants, current sensor or encoder shall be treated as a controlled engineering change; approval requires FOC model correlation, parameter sweep and sensor-fault testing.	FOC model correlation, parameter sweep and sensor-fault testing

Verification, margin and future-change rule

Changes involving motor, FOC algorithm, motor constants, current sensor or encoder are engineering events, not editorial updates; approval depends on the targeted FOC model correlation, parameter sweep and sensor-fault testing listed here.

50 kHz+ CURRENT-LOOP EXECUTION

The current loop shall execute at the baseline 50 kHz+ update rate with a controlled worst-case latency and jitter budget.

If deadline overrun or timing jitter exceeds the approved bound, the system shall flag the timing fault and prevent continued high-authority motion unless the safety case explicitly permits degradation without relying on an unrelated convenience function.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-028	The current loop shall execute at the baseline 50 kHz+ update rate with a controlled worst-case latency and jitter budget.	WCET/timing measurement, stress test and deadline fault injection
RSMC50K-029	Loss or corruption involving deadline overrun or timing jitter exceeds the approved bound shall not permit silent normal operation; the system shall flag the timing fault and prevent continued high-authority motion unless the safety case explicitly permits degradation.	WCET/timing measurement, stress test and deadline fault injection
RSMC50K-030	Production equivalence following FPGA clock, loop rate, code path, arithmetic width or resource utilization requires closure of WCET/timing measurement, stress test and deadline fault injection.	WCET/timing measurement, stress test and deadline fault injection

Verification, margin and future-change rule

For FPGA clock, loop rate, code path, arithmetic width or resource utilization, the configuration owner documents evidence reuse, new risks and the required WCET/timing measurement, stress test and deadline fault injection before sign-off.

PWM, DEAD TIME & GATE-TIMING CONTROL

PWM generation shall maintain controlled dead time, minimum pulse behavior and synchronized sampling across all released operating modes.

Pwm timing becomes inconsistent or a complementary gate conflict is detected is treated as a diagnosable abnormal condition. The system response is to force the affected bridge leg to the validated safe state and latch diagnostics.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-031	PWM generation shall maintain controlled dead time, minimum pulse behavior and synchronized sampling across all released operating modes.	timing analysis, gate waveform test and injected overlap fault
RSMC50K-032	When PWM timing becomes inconsistent or a complementary gate conflict is detected, the product shall force the affected bridge leg to the validated safe state and latch diagnostics.	timing analysis, gate waveform test and injected overlap fault
RSMC50K-033	The released configuration shall not be extended across PWM architecture, gate driver, dead time, switching frequency or modulation without timing analysis, gate waveform test and injected overlap fault.	timing analysis, gate waveform test and injected overlap fault

Verification, margin and future-change rule

Any modification to PWM architecture, gate driver, dead time, switching frequency or modulation is accepted only after the affected assumptions are rechecked and timing analysis, gate waveform test and injected overlap fault is closed.

POSITION-LOOP CONTROL & SUB-MILLIMETER MISSION

Position-control behavior shall support the baseline sub-millimeter precision mission within a program-defined mechanical, sensor and calibration envelope.

The design prevents silent continuation when position error or estimator inconsistency exceeds the controlled supervision threshold; it must instead limit or stop commanded motion according to the allocated safety response.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-034	Position-control behavior shall support the baseline sub-millimeter precision mission within a program-defined mechanical, sensor and calibration envelope.	precision DVP, trajectory test and error-supervision fault injection
RSMC50K-035	The released implementation shall limit or stop commanded motion according to the allocated safety response whenever position error or estimator inconsistency exceeds the controlled supervision threshold.	precision DVP, trajectory test and error-supervision fault injection
RSMC50K-036	Changes to encoder, mechanics, control gains, trajectory generator or calibration shall trigger impact review of affected assumptions and closure of precision DVP, trajectory test and error-supervision fault injection.	precision DVP, trajectory test and error-supervision fault injection

Verification, margin and future-change rule

The release baseline remains valid across encoder, mechanics, control gains, trajectory generator or calibration only within demonstrated similarity. Otherwise, precision DVP, trajectory test and error-supervision fault injection is repeated or updated.

FORCE / TORQUE CONTROL & CONTACT BEHAVIOR

Force/torque control shall use validated sensing or estimation and remain subordinate to independent hardware-enforced limits.

If force estimate is invalid, contradictory or outside the allowed control envelope, the system shall remove or limit torque before continued motion can exceed the safety allocation.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-037	Force/torque control shall use validated sensing or estimation and remain subordinate to independent hardware-enforced limits.	force-control correlation, contact test and sensor/estimator fault injection
RSMC50K-038	If force estimate is invalid, contradictory or outside the allowed control envelope, the system shall remove or limit torque before continued motion can exceed the safety allocation.	force-control correlation, contact test and sensor/estimator fault injection
RSMC50K-039	A change to force sensor, torque estimator, motor model, control gain or mechanical transmission shall reopen force-control correlation, contact test and sensor/estimator fault injection before production release.	force-control correlation, contact test and sensor/estimator fault injection

Verification, margin and future-change rule

Future variants affecting force sensor, torque estimator, motor model, control gain or mechanical transmission are screened against the mechanism on this page. The release owner may retain prior evidence only where force-control correlation, contact test and sensor/estimator fault injection confirms the similarity boundary.

KINEMATIC TRANSFORM, DETERMINISTIC TIMING & JITTER BOUND

The VHDL/C kinematic transform path shall have a measured deterministic execution bound; the baseline phrase zero-jitter shall be converted to an approved numerical jitter limit before final release.

If kinematic data arrives late, out of order or exceeds the released timing bound, the product shall reject the invalid command frame and hold or stop motion per the state machine.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-040	The VHDL/C kinematic transform path shall have a measured deterministic execution bound; the baseline phrase zero-jitter shall be converted to an approved numerical jitter limit before final release.	timing trace, WCET measurement and stale/out-of-order command tests
RSMC50K-041	If kinematic data arrives late, out of order or exceeds the released timing bound, the system shall reject the invalid command frame and hold or stop motion per the state machine.	timing trace, WCET measurement and stale/out-of-order command tests
RSMC50K-042	Evidence reuse after kinematic algorithm, clocking, bus protocol, arithmetic implementation or scheduling is permitted only when timing trace, WCET measurement and stale/out-of-order command tests demonstrates that the affected assumptions remain valid.	timing trace, WCET measurement and stale/out-of-order command tests

Verification, margin and future-change rule

The change trigger for this page is kinematic algorithm, clocking, bus protocol, arithmetic implementation or scheduling. The impact record identifies moved assumptions and the exact timing trace, WCET measurement and stale/out-of-order command tests needed for closure.

MOTION COMMAND INTERFACE, FRESHNESS & RANGE CHECKS

Incoming motion commands shall carry valid identity, sequence, freshness and range information before they are accepted by the servo path.

Verification focuses on the point where command data is stale, duplicated, malformed or outside the approved workspace. The required containment is to reject the command and preserve the last safe or zero-motion authority as allocated.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-043	Incoming motion commands shall carry valid identity, sequence, freshness and range information before they are accepted by the servo path.	interface conformance, freshness tests and malformed-command injection
RSMC50K-044	The design shall detect command data is stale, duplicated, malformed or outside the approved workspace and shall reject the command and preserve the last safe or zero-motion authority as allocated.	interface conformance, freshness tests and malformed-command injection
RSMC50K-045	Host protocol, command rate, sequence counter, workspace definition or backplane shall be treated as a controlled engineering change; approval requires interface conformance, freshness tests and malformed-command injection.	interface conformance, freshness tests and malformed-command injection

Verification, margin and future-change rule

Changes involving host protocol, command rate, sequence counter, workspace definition or backplane are engineering events, not editorial updates; approval depends on the targeted interface conformance, freshness tests and malformed-command injection listed here.

COLLISION / ABNORMAL CONTACT DETECTION

The controller shall detect allocated abnormal contact or collision indicators using the program-defined sensor/estimator combination.

If contact metrics indicate an unsafe or unclassified collision condition, the system shall invoke force limiting or STO/controlled stop according to the validated risk-control timing without relying on an unrelated convenience function.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-046	The controller shall detect allocated abnormal contact or collision indicators using the program-defined sensor/estimator combination.	collision DVP, contact-force test and fault timing analysis
RSMC50K-047	If contact metrics indicate an unsafe or unclassified collision condition, the system shall invoke force limiting or STO/controlled stop according to the validated risk-control timing.	collision DVP, contact-force test and fault timing analysis
RSMC50K-048	Production equivalence following collision algorithm, sensor set, thresholds, mechanical tool or robot dynamics requires closure of collision DVP, contact-force test and fault timing analysis.	collision DVP, contact-force test and fault timing analysis

Verification, margin and future-change rule

For collision algorithm, sensor set, thresholds, mechanical tool or robot dynamics, the configuration owner documents evidence reuse, new risks and the required collision DVP, contact-force test and fault timing analysis before sign-off.

HARDWARE-ENFORCED FORCE / TORQUE LIMITS

Independent hardware-enforced force or torque limits shall be capable of constraining hazardous output without relying solely on high-level motion software.

Measured or derived force/torque exceeds the hardware protection threshold is treated as a diagnosable abnormal condition. The system response is to remove or clamp drive authority within the controlled protection-time budget.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-049	Independent hardware-enforced force or torque limits shall be capable of constraining hazardous output without relying solely on high-level motion software.	hardware limit test, threshold tolerance analysis and independence review
RSMC50K-050	When measured or derived force/torque exceeds the hardware protection threshold, the product shall remove or clamp drive authority within the controlled protection-time budget.	hardware limit test, threshold tolerance analysis and independence review
RSMC50K-051	The released configuration shall not be extended across limit circuit, current sensor, comparator, threshold component or power stage without hardware limit test, threshold tolerance analysis and independence review.	hardware limit test, threshold tolerance analysis and independence review

Verification, margin and future-change rule

Any modification to limit circuit, current sensor, comparator, threshold component or power stage is accepted only after the affected assumptions are rechecked and hardware limit test, threshold tolerance analysis and independence review is closed.

SAFE TORQUE OFF (STO) INDEPENDENT PATH

STO shall remove torque-producing gate authority through the released independent hardware path and require controlled re-enable conditions.

The design prevents silent continuation when STO is demanded or its monitoring detects an unsafe discrepancy; it must instead de-energize torque authority and prevent restart until the defined reset/revalidation sequence is complete.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-052	STO shall remove torque-producing gate authority through the released independent hardware path and require controlled re-enable conditions.	STO timing test, discrepancy fault injection and restart validation
RSMC50K-053	The released implementation shall de-energize torque authority and prevent restart until the defined reset/revalidation sequence is complete whenever STO is demanded or its monitoring detects an unsafe discrepancy.	STO timing test, discrepancy fault injection and restart validation
RSMC50K-054	Changes to STO input, gate driver, enable chain, monitor circuit or external safety interface shall trigger impact review of affected assumptions and closure of STO timing test, discrepancy fault injection and restart validation.	STO timing test, discrepancy fault injection and restart validation

Verification, margin and future-change rule

The release baseline remains valid across STO input, gate driver, enable chain, monitor circuit or external safety interface only within demonstrated similarity. Otherwise, STO timing test, discrepancy fault injection and restart validation is repeated or updated.

CONTROLLED STOP, BRAKE & HOLDING INTERACTION

Controlled stop behavior shall coordinate torque ramp-down, brake engagement and holding-state confirmation without delaying independent STO.

If brake feedback, stop trajectory or holding state is invalid, the system shall default to the allocated safe strategy and preserve STO availability.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-055	Controlled stop behavior shall coordinate torque ramp-down, brake engagement and holding-state confirmation without delaying independent STO.	stop/brake timing DVP, feedback fault injection and gravity/load test
RSMC50K-056	If brake feedback, stop trajectory or holding state is invalid, the system shall default to the allocated safe strategy and preserve STO availability.	stop/brake timing DVP, feedback fault injection and gravity/load test
RSMC50K-057	A change to brake, holding mechanism, stop profile, load, orientation or feedback sensor shall reopen stop/brake timing DVP, feedback fault injection and gravity/load test before production release.	stop/brake timing DVP, feedback fault injection and gravity/load test

Verification, margin and future-change rule

Future variants affecting brake, holding mechanism, stop profile, load, orientation or feedback sensor are screened against the mechanism on this page. The release owner may retain prior evidence only where stop/brake timing DVP, feedback fault injection and gravity/load test confirms the similarity boundary.

FAULT CONTAINMENT, LATCHING & RE-ENERGIZATION

Safety-relevant faults shall have defined latching, acknowledgement and re-energization criteria so transient recovery cannot create unexpected motion.

If a latched motion fault clears without the required external or internal validation, the product shall keep torque inhibited and record the failed reset precondition.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-058	Safety-relevant faults shall have defined latching, acknowledgement and re-energization criteria so transient recovery cannot create unexpected motion.	fault-reset test, power-cycle test and unexpected re-enable injection
RSMC50K-059	If a latched motion fault clears without the required external or internal validation, the system shall keep torque inhibited and record the failed reset precondition.	fault-reset test, power-cycle test and unexpected re-enable injection
RSMC50K-060	Evidence reuse after fault manager, reset policy, power cycle, host acknowledgement or service function is permitted only when fault-reset test, power-cycle test and unexpected re-enable injection demonstrates that the affected assumptions remain valid.	fault-reset test, power-cycle test and unexpected re-enable injection

Verification, margin and future-change rule

The change trigger for this page is fault manager, reset policy, power cycle, host acknowledgement or service function. The impact record identifies moved assumptions and the exact fault-reset test, power-cycle test and unexpected re-enable injection needed for closure.

WATCHDOG, CLOCK, RESET & POWER-ON SUPERVISION

Independent supervision shall detect control-loop stall, clock failure, reset anomaly and unsafe startup sequencing.

Verification focuses on the point where watchdog, clock or reset supervision reports loss of valid execution. The required containment is to remove motion authority through the fastest allocated path and capture first-fault status.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-061	Independent supervision shall detect control-loop stall, clock failure, reset anomaly and unsafe startup sequencing.	watchdog/clock fault injection, reset sequencing and startup DVP
RSMC50K-062	The design shall detect watchdog, clock or reset supervision reports loss of valid execution and shall remove motion authority through the fastest allocated path and capture first-fault status.	watchdog/clock fault injection, reset sequencing and startup DVP
RSMC50K-063	Watchdog, oscillator, FPGA clock, reset IC or startup firmware shall be treated as a controlled engineering change; approval requires watchdog/clock fault injection, reset sequencing and startup DVP.	watchdog/clock fault injection, reset sequencing and startup DVP

Verification, margin and future-change rule

Changes involving watchdog, oscillator, FPGA clock, reset IC or startup firmware are engineering events, not editorial updates; approval depends on the targeted watchdog/clock fault injection, reset sequencing and startup DVP listed here.

FPGA CONFIGURATION INTEGRITY & STARTUP AUTHENTICATION

FPGA configuration shall be protected against corruption and shall not enable motor outputs until the released image and startup diagnostics are valid.

If configuration CRC/authenticity or startup self-test fails, the system shall hold gate authority disabled and enter a diagnosable recovery/service state without relying on an unrelated convenience function.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-064	FPGA configuration shall be protected against corruption and shall not enable motor outputs until the released image and startup diagnostics are valid.	configuration corruption tests, startup test and image-control review
RSMC50K-065	Loss or corruption involving configuration CRC/authenticity or startup self-test fails shall not permit silent normal operation; the system shall hold gate authority disabled and enter a diagnosable recovery/service state.	configuration corruption tests, startup test and image-control review
RSMC50K-066	Production equivalence following FPGA image, bitstream storage, secure boot, configuration memory or startup logic requires closure of configuration corruption tests, startup test and image-control review.	configuration corruption tests, startup test and image-control review

Verification, margin and future-change rule

For FPGA image, bitstream storage, secure boot, configuration memory or startup logic, the configuration owner documents evidence reuse, new risks and the required configuration corruption tests, startup test and image-control review before sign-off.

BARE-METAL VHDL / C LIFECYCLE & CONFIGURATION CONTROL

VHDL/C logic shall be traceable to controlled requirements, implementation, verification results and exact toolchain/build configuration.

Build identity or verification traceability is incomplete is treated as a diagnosable abnormal condition. The system response is to block release of the affected software/FPGA image rather than compensating at runtime.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-067	VHDL/C logic shall be traceable to controlled requirements, implementation, verification results and exact toolchain/build configuration.	configuration audit, traceability review and reproducible-build evidence
RSMC50K-068	If build identity or verification traceability is incomplete, release of the affected software/FPGA image shall be blocked rather than compensated at runtime.	configuration audit, traceability review and reproducible-build evidence
RSMC50K-069	The released configuration shall not be extended across compiler, synthesis tool, source revision, library or build option without configuration audit, traceability review and reproducible-build evidence.	configuration audit, traceability review and reproducible-build evidence

Verification, margin and future-change rule

Any modification to compiler, synthesis tool, source revision, library or build option is accepted only after the affected assumptions are rechecked and configuration audit, traceability review and reproducible-build evidence is closed.

DETERMINISTIC INTERNAL BACKPLANE

The hardwired internal backplane shall provide bounded latency, sequence integrity and fault containment for motion/safety exchanges.

The design prevents silent continuation when frame loss, corruption, timeout or sequence error occurs on the backplane; it must instead invalidate the affected command/status and transition to the defined limited or safe motion state.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-070	The hardwired internal backplane shall provide bounded latency, sequence integrity and fault containment for motion/safety exchanges.	bus timing DVP, error injection and load testing
RSMC50K-071	The released implementation shall invalidate the affected command/status and transition to the defined limited or safe motion state whenever frame loss, corruption, timeout or sequence error occurs on the backplane.	bus timing DVP, error injection and load testing
RSMC50K-072	Changes to backplane PHY, protocol, cable, connector, frame rate or topology shall trigger impact review of affected assumptions and closure of bus timing DVP, error injection and load testing.	bus timing DVP, error injection and load testing

Verification, margin and future-change rule

The release baseline remains valid across backplane PHY, protocol, cable, connector, frame rate or topology only within demonstrated similarity. Otherwise, bus timing DVP, error injection and load testing is repeated or updated.

ISOLATION FROM NON-SURGICAL INTERFACES

Non-surgical service, logging or hospital-network interfaces shall not gain direct real-time motion or STO authority.

If a non-surgical interface attempts to inject or alter a motion-critical message, the system shall block the path at the partition boundary and record the event.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-073	Non-surgical service, logging or hospital-network interfaces shall not gain direct real-time motion or STO authority.	architecture review, negative routing tests and penetration/fault injection
RSMC50K-074	If a non-surgical interface attempts to inject or alter a motion-critical message, the system shall block the path at the partition boundary and record the event.	architecture review, negative routing tests and penetration/fault injection
RSMC50K-075	A change to service port, diagnostic bridge, external network, logging path or software partition shall reopen architecture review, negative routing tests and penetration/fault injection before production release.	architecture review, negative routing tests and penetration/fault injection

Verification, margin and future-change rule

Future variants affecting service port, diagnostic bridge, external network, logging path or software partition are screened against the mechanism on this page. The release owner may retain prior evidence only where architecture review, negative routing tests and penetration/fault injection confirms the similarity boundary.

CYBERSECURITY, MAINTENANCE ACCESS & TRUSTED SERVICE

Maintenance access shall be authenticated and constrained to approved calibration, diagnostics or update actions without bypassing local safety protections.

If service authentication fails or an unauthorized privileged action is requested, the product shall deny the action, preserve safe motion state and log the attempt.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-076	Maintenance access shall be authenticated and constrained to approved calibration, diagnostics or update actions without bypassing local safety protections.	service-access test, role/permission review and negative authorization cases
RSMC50K-077	If service authentication fails or an unauthorized privileged action is requested, the system shall deny the action, preserve safe motion state and log the attempt.	service-access test, role/permission review and negative authorization cases
RSMC50K-078	Evidence reuse after service tool, credential policy, debug port, maintenance protocol or role model is permitted only when service-access test, role/permission review and negative authorization cases demonstrates that the affected assumptions remain valid.	service-access test, role/permission review and negative authorization cases

Verification, margin and future-change rule

The change trigger for this page is service tool, credential policy, debug port, maintenance protocol or role model. The impact record identifies moved assumptions and the exact service-access test, role/permission review and negative authorization cases needed for closure.

SECURE UPDATE, IMAGE COMPATIBILITY & ROLLBACK

Firmware/FPGA updates shall verify authenticity, hardware compatibility and rollback/recovery behavior before motion enable.

Verification focuses on the point where update validation or post-installation self-test fails. The required containment is to retain or restore the last approved image and keep motion disabled until recovery is complete.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-079	Firmware/FPGA updates shall verify authenticity, hardware compatibility and rollback/recovery behavior before motion enable.	interrupted-update test, compatibility test and rollback verification
RSMC50K-080	The design shall detect update validation or post-installation self-test fails and shall retain or restore the last approved image and keep motion disabled until recovery is complete.	interrupted-update test, compatibility test and rollback verification
RSMC50K-081	Update package, signing key, bootloader, FPGA image or hardware revision shall be treated as a controlled engineering change; approval requires interrupted-update test, compatibility test and rollback verification.	interrupted-update test, compatibility test and rollback verification

Verification, margin and future-change rule

Changes involving update package, signing key, bootloader, FPGA image or hardware revision are engineering events, not editorial updates; approval depends on the targeted interrupted-update test, compatibility test and rollback verification listed here.

CALIBRATION AUTHORITY, PARAMETER BOUNDS & VARIANT CODING

Motor, encoder, force, friction and control parameters shall be range-checked and bound to the correct hardware/mechanical variant.

If calibration identity, checksum or variant compatibility is invalid, the system shall prevent normal motion enable and require authorized recalibration or service disposition without relying on an unrelated convenience function.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-082	Motor, encoder, force, friction and control parameters shall be range-checked and bound to the correct hardware/mechanical variant.	calibration CRC/range test, variant mismatch test and traceability review
RSMC50K-083	If calibration identity, checksum or variant compatibility is invalid, the system shall prevent normal motion enable and require authorized recalibration or service disposition.	calibration CRC/range test, variant mismatch test and traceability review
RSMC50K-084	Production equivalence following calibration dataset, motor variant, gear ratio, encoder, tool or controller gains requires closure of calibration CRC/range test, variant mismatch test and traceability review.	calibration CRC/range test, variant mismatch test and traceability review

Verification, margin and future-change rule

For calibration dataset, motor variant, gear ratio, encoder, tool or controller gains, the configuration owner documents evidence reuse, new risks and the required calibration CRC/range test, variant mismatch test and traceability review before sign-off.

ENCODER OFFSET CALIBRATION

Manufacturing/service calibration shall establish encoder electrical/mechanical offset with traceable acceptance criteria.

Encoder offset cannot be calibrated within the released acceptance window is treated as a diagnosable abnormal condition. The system response is to reject or service the affected joint and prevent precision-motion release.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-085	Manufacturing/service calibration shall establish encoder electrical/mechanical offset with traceable acceptance criteria.	encoder calibration procedure, repeatability study and EOL records
RSMC50K-086	When encoder offset cannot be calibrated within the released acceptance window, the product shall reject or service the affected joint and prevent precision-motion release.	encoder calibration procedure, repeatability study and EOL records
RSMC50K-087	The released configuration shall not be extended across encoder, mounting stack, rotor alignment, calibration fixture or algorithm without encoder calibration procedure, repeatability study and EOL records.	encoder calibration procedure, repeatability study and EOL records

Verification, margin and future-change rule

Any modification to encoder, mounting stack, rotor alignment, calibration fixture or algorithm is accepted only after the affected assumptions are rechecked and encoder calibration procedure, repeatability study and EOL records is closed.

STATIC FRICTION / COGGING MAPPING

The controller shall use approved friction/cogging compensation maps only within the conditions and mechanism for which they were validated.

The design prevents silent continuation when friction map is missing, stale or inconsistent with the mechanical variant; it must instead disable the compensation or use the validated fallback while preserving force-limit authority.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-088	The controller shall use approved friction/cogging compensation maps only within the conditions and mechanism for which they were validated.	friction-map correlation, variant test and disabled-map comparison
RSMC50K-089	The released implementation shall disable the compensation or use the validated fallback while preserving force-limit authority whenever friction map is missing, stale or inconsistent with the mechanical variant.	friction-map correlation, variant test and disabled-map comparison
RSMC50K-090	Changes to gearbox, lubrication, motor, compensation map, temperature or mechanical preload shall trigger impact review of affected assumptions and closure of friction-map correlation, variant test and disabled-map comparison.	friction-map correlation, variant test and disabled-map comparison

Verification, margin and future-change rule

The release baseline remains valid across gearbox, lubrication, motor, compensation map, temperature or mechanical preload only within demonstrated similarity. Otherwise, friction-map correlation, variant test and disabled-map comparison is repeated or updated.

FORCE / TORQUE CALIBRATION & CROSS-CORRELATION

Force/torque sensing or estimation shall be calibrated against traceable references and cross-checked with motor-current/position behavior where allocated.

If force calibration drift or cross-correlation falls outside acceptance limits, the system shall inhibit high-authority surgical motion and require recalibration or service.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-091	Force/torque sensing or estimation shall be calibrated against traceable references and cross-checked with motor-current/position behavior where allocated.	force calibration, correlation test and drift/offset fault injection
RSMC50K-092	If force calibration drift or cross-correlation falls outside acceptance limits, the system shall inhibit high-authority surgical motion and require recalibration or service.	force calibration, correlation test and drift/offset fault injection
RSMC50K-093	A change to force sensor, current sensor, fixture, estimator model or calibration coefficients shall reopen force calibration, correlation test and drift/offset fault injection before production release.	force calibration, correlation test and drift/offset fault injection

Verification, margin and future-change rule

Future variants affecting force sensor, current sensor, fixture, estimator model or calibration coefficients are screened against the mechanism on this page. The release owner may retain prior evidence only where force calibration, correlation test and drift/offset fault injection confirms the similarity boundary.

CONDUCTION COOLING, JUNCTION / PCB TEMPERATURE & DERATING

Joint-mounted conduction cooling shall control semiconductor and electronics temperature without active fan dependence.

If temperature exceeds the released derating or shutdown threshold, or sensor validity is lost, the product shall limit motion capability or remove torque according to the thermal protection allocation.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-094	Joint-mounted conduction cooling shall control semiconductor and electronics temperature without active fan dependence.	thermal model, worst-case motion test and sensor-fault injection
RSMC50K-095	If temperature exceeds the released derating or shutdown threshold, or sensor validity is lost, the system shall limit motion capability or remove torque according to the thermal protection allocation.	thermal model, worst-case motion test and sensor-fault injection
RSMC50K-096	Evidence reuse after heat spreader, TIM, power device, joint housing, ambient or motion duty cycle is permitted only when thermal model, worst-case motion test and sensor-fault injection demonstrates that the affected assumptions remain valid.	thermal model, worst-case motion test and sensor-fault injection

Verification, margin and future-change rule

The change trigger for this page is heat spreader, TIM, power device, joint housing, ambient or motion duty cycle. The impact record identifies moved assumptions and the exact thermal model, worst-case motion test and sensor-fault injection needed for closure.

ACOUSTIC CONTROL & NO-FAN DESIGN INTENT

The controller shall meet the operating-room acoustic objective through passive/conduction cooling; the baseline phrase 'zero acoustic noise' shall be converted to an approved measurable criterion before release.

Verification focuses on the point where an added fan, switching pattern or mechanical resonance creates unapproved audible behavior. The required containment is to prevent release of the affected variant until the acoustic requirement and evidence are closed.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-097	The controller shall meet the operating-room acoustic objective through passive/conduction cooling; the baseline phrase 'zero acoustic noise' shall be converted to an approved measurable criterion before release.	acoustic measurement, switching-tone review and thermal/acoustic trade study
RSMC50K-098	The design shall detect an added fan, switching pattern or mechanical resonance creates unapproved audible behavior and shall prevent release of the affected variant until the acoustic requirement and evidence are closed.	acoustic measurement, switching-tone review and thermal/acoustic trade study
RSMC50K-099	Cooling method, switching frequency, enclosure, fan, coil or mechanical mounting shall be treated as a controlled engineering change; approval requires acoustic measurement, switching-tone review and thermal/acoustic trade study.	acoustic measurement, switching-tone review and thermal/acoustic trade study

Verification, margin and future-change rule

Changes involving cooling method, switching frequency, enclosure, fan, coil or mechanical mounting are engineering events, not editorial updates; approval depends on the targeted acoustic measurement, switching-tone review and thermal/acoustic trade study listed here.

EMI SHIELDING, EMC & SIGNAL INTEGRITY

The shielded joint module shall maintain current/encoder/backplane integrity across the approved medical EMC environment.

If EMI causes false encoder counts, current-sense corruption, reset or safety-channel disturbance, the system shall detect invalid data or transition to the allocated safe state without uncontrolled motion without relying on an unrelated convenience function.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-100	The shielded joint module shall maintain current/encoder/backplane integrity across the approved medical EMC environment.	EMC/ESD test, signal-integrity analysis and post-stress functional DVP
RSMC50K-101	Loss or corruption involving EMI causes false encoder counts, current-sense corruption, reset or safety-channel disturbance shall not permit silent normal operation; the system shall detect invalid data or transition to the allocated safe state without uncontrolled motion.	EMC/ESD test, signal-integrity analysis and post-stress functional DVP
RSMC50K-102	Production equivalence following shielding, cable, connector, filter, PCB layout or enclosure requires closure of EMC/ESD test, signal-integrity analysis and post-stress functional DVP.	EMC/ESD test, signal-integrity analysis and post-stress functional DVP

Verification, margin and future-change rule

For shielding, cable, connector, filter, PCB layout or enclosure, the configuration owner documents evidence reuse, new risks and the required EMC/ESD test, signal-integrity analysis and post-stress functional DVP before sign-off.

STERILIZATION / CLEANING EXPOSURE & MATERIAL COMPATIBILITY

Materials, seals, labels and electronics interfaces shall tolerate the approved sterilization or cleaning process allocated to the joint-mounted module.

Exposure changes insulation, mechanical fit, calibration or sealing beyond accepted limits is treated as a diagnosable abnormal condition. The system response is to remove the affected configuration from release/service until requalification is complete.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-103	Materials, seals, labels and electronics interfaces shall tolerate the approved sterilization or cleaning process allocated to the joint-mounted module.	material compatibility, repeated-exposure test and post-cycle calibration/safety check
RSMC50K-104	When exposure changes insulation, mechanical fit, calibration or sealing beyond accepted limits, the product shall remove the affected configuration from release/service until requalification is complete.	material compatibility, repeated-exposure test and post-cycle calibration/safety check
RSMC50K-105	The released configuration shall not be extended across sterilization method, cleaning agent, polymer, coating, seal or adhesive without material compatibility, repeated-exposure test and post-cycle calibration/safety check.	material compatibility, repeated-exposure test and post-cycle calibration/safety check

Verification, margin and future-change rule

Any modification to sterilization method, cleaning agent, polymer, coating, seal or adhesive is accepted only after the affected assumptions are rechecked and material compatibility, repeated-exposure test and post-cycle calibration/safety check is closed.

JOINT-MOUNTED MECHANICAL INTEGRATION, SHOCK & STRAIN

The module shall maintain connector, sensor, thermal and grounding interfaces under the released robotic-joint mechanical loads.

The design prevents silent continuation when shock, strain or fastener movement changes critical electrical or calibration geometry; it must instead enter service/fault state as detectable and prevent unqualified precision operation.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-106	The module shall maintain connector, sensor, thermal and grounding interfaces under the released robotic-joint mechanical loads.	mechanical analysis, shock/vibration test and post-stress calibration
RSMC50K-107	The released implementation shall enter service/fault state as detectable and prevent unqualified precision operation whenever shock, strain or fastener movement changes critical electrical or calibration geometry.	mechanical analysis, shock/vibration test and post-stress calibration
RSMC50K-108	Changes to joint bracket, fastener, harness, enclosure, connector or thermal stack shall trigger impact review of affected assumptions and closure of mechanical analysis, shock/vibration test and post-stress calibration.	mechanical analysis, shock/vibration test and post-stress calibration

Verification, margin and future-change rule

The release baseline remains valid across joint bracket, fastener, harness, enclosure, connector or thermal stack only within demonstrated similarity. Otherwise, mechanical analysis, shock/vibration test and post-stress calibration is repeated or updated.

RELIABILITY, MOTION-CYCLE MISSION & WEAR INDICATORS

Reliability evidence shall reflect the approved motion-cycle, load, temperature and sterilization/cleaning mission rather than generic calendar time alone.

If wear indicators or accumulated use exceed the service threshold, the system shall flag maintenance/replacement status before degraded mechanics can compromise precision or safety.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-109	Reliability evidence shall reflect the approved motion-cycle, load, temperature and sterilization/cleaning mission rather than generic calendar time alone.	mission-profile analysis, endurance test and service-threshold review
RSMC50K-110	If wear indicators or accumulated use exceed the service threshold, the system shall flag maintenance/replacement status before degraded mechanics can compromise precision or safety.	mission-profile analysis, endurance test and service-threshold review
RSMC50K-111	A change to duty cycle, load, temperature, gearbox, bearing or service interval shall reopen mission-profile analysis, endurance test and service-threshold review before production release.	mission-profile analysis, endurance test and service-threshold review

Verification, margin and future-change rule

Future variants affecting duty cycle, load, temperature, gearbox, bearing or service interval are screened against the mechanism on this page. The release owner may retain prior evidence only where mission-profile analysis, endurance test and service-threshold review confirms the similarity boundary.

GEARBOX-WEAR MONITORING BY MOTOR-CURRENT SIGNATURE

Current-signature monitoring shall trend gearbox wear only within validated speed/load/temperature regions and shall expose uncertainty outside them.

If wear metric quality is invalid or exceeds the maintenance threshold, the product shall issue predictive-maintenance status without overriding local force/STO safety controls.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-112	Current-signature monitoring shall trend gearbox wear only within validated speed/load/temperature regions and shall expose uncertainty outside them.	wear-correlation test, seeded-fault evaluation and field-data review
RSMC50K-113	If wear metric quality is invalid or exceeds the maintenance threshold, the system shall issue predictive-maintenance status without overriding local force/STO safety controls.	wear-correlation test, seeded-fault evaluation and field-data review
RSMC50K-114	Evidence reuse after gearbox, current sensor, feature algorithm, threshold or operating profile is permitted only when wear-correlation test, seeded-fault evaluation and field-data review demonstrates that the affected assumptions remain valid.	wear-correlation test, seeded-fault evaluation and field-data review

Verification, margin and future-change rule

The change trigger for this page is gearbox, current sensor, feature algorithm, threshold or operating profile. The impact record identifies moved assumptions and the exact wear-correlation test, seeded-fault evaluation and field-data review needed for closure.

MANUFACTURING PROGRAMMING, GENEALOGY & CONFIGURATION IDENTITY

Each controller shall be traceable to FPGA/software image, power-stage, sensors, calibration and manufacturing test records.

Verification focuses on the point where genealogy or image/calibration identity is incomplete. The required containment is to block unit release until the record is reconciled and the exact configuration is known.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-115	Each controller shall be traceable to FPGA/software image, power-stage, sensors, calibration and manufacturing test records.	MES genealogy audit, serialization test and configuration reconciliation
RSMC50K-116	The design shall detect genealogy or image/calibration identity is incomplete and shall block unit release until the record is reconciled and the exact configuration is known.	MES genealogy audit, serialization test and configuration reconciliation
RSMC50K-117	BOM, FPGA image, calibration, manufacturing site or test database shall be treated as a controlled engineering change; approval requires MES genealogy audit, serialization test and configuration reconciliation.	MES genealogy audit, serialization test and configuration reconciliation

Verification, margin and future-change rule

Changes involving BOM, FPGA image, calibration, manufacturing site or test database are engineering events, not editorial updates; approval depends on the targeted MES genealogy audit, serialization test and configuration reconciliation listed here.

EOL ELECTRICAL, I/O & STO FUNCTIONAL TEST

EOL shall verify power rails, current/encoder channels, backplane communication and independent STO actuation for the released hardware.

If required EOL functional evidence is missing or outside limits, the system shall reject or route the unit for controlled rework before release without relying on an unrelated convenience function.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-118	EOL shall verify power rails, current/encoder channels, backplane communication and independent STO actuation for the released hardware.	EOL test records, measurement-system analysis and fault-path coverage
RSMC50K-119	Loss or corruption involving required EOL functional evidence is missing or outside limits shall not permit silent normal operation; the system shall reject or route the unit for controlled rework before release.	EOL test records, measurement-system analysis and fault-path coverage
RSMC50K-120	Production equivalence following test fixture, hardware revision, STO path, sensor channel or EOL limit requires closure of EOL test records, measurement-system analysis and fault-path coverage.	EOL test records, measurement-system analysis and fault-path coverage

Verification, margin and future-change rule

For test fixture, hardware revision, STO path, sensor channel or EOL limit, the configuration owner documents evidence reuse, new risks and the required EOL test records, measurement-system analysis and fault-path coverage before sign-off.

EOL PRECISION CALIBRATION & FRICTION MAP ACCEPTANCE

EOL shall close encoder offset, friction mapping and force/torque calibration evidence for the exact mechanical variant.

Calibration residual, repeatability or variant identity is outside acceptance criteria is treated as a diagnosable abnormal condition. The system response is to withhold precision-motion release until corrected and reverified.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-121	EOL shall close encoder offset, friction mapping and force/torque calibration evidence for the exact mechanical variant.	calibration reports, repeatability/capability study and variant traceability
RSMC50K-122	If calibration residual, repeatability or variant identity is outside acceptance criteria, precision-motion release shall be withheld until corrected and reverified.	calibration reports, repeatability/capability study and variant traceability
RSMC50K-123	The released configuration shall not be extended across fixture, calibration algorithm, gearbox, motor, encoder or force sensor without calibration reports, repeatability/capability study and variant traceability.	calibration reports, repeatability/capability study and variant traceability

Verification, margin and future-change rule

Any modification to fixture, calibration algorithm, gearbox, motor, encoder or force sensor is accepted only after the affected assumptions are rechecked and calibration reports, repeatability/capability study and variant traceability is closed.

PRECISION, LATENCY & CLOSED-LOOP DVP

DVP shall demonstrate sub-millimeter mission performance, 50 kHz+ current-loop timing and closed-loop stability across the approved load, trajectory and thermal envelope.

The design prevents silent continuation when a required motion envelope, acceptance limit or timing criterion is not closed; it must instead keep the release gate open rather than inferring production capability from nominal bench tests.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-124	DVP shall demonstrate sub-millimeter mission performance, 50 kHz+ current-loop timing and closed-loop stability across the approved load, trajectory and thermal envelope.	trajectory/precision DVP, timing traces and stability margins
RSMC50K-125	The released implementation shall keep the release gate open rather than inferring production capability from nominal bench tests whenever a required motion envelope, acceptance limit or timing criterion is not closed.	trajectory/precision DVP, timing traces and stability margins
RSMC50K-126	Changes to motion envelope, payload, trajectory, thermal condition, control gain or timing criterion shall trigger impact review of affected assumptions and closure of trajectory/precision DVP, timing traces and stability margins.	trajectory/precision DVP, timing traces and stability margins

Verification, margin and future-change rule

The release baseline remains valid across motion envelope, payload, trajectory, thermal condition, control gain or timing criterion only within demonstrated similarity. Otherwise, trajectory/precision DVP, timing traces and stability margins is repeated or updated.

FAULT-INJECTION, FORCE-LIMIT & STO DVP

Verification shall inject sensor, FPGA/software, backplane, power and gate-path faults while measuring force-limit and STO response.

If fault injection reveals delayed, missing or ambiguous torque removal, the system shall fail the safety release gate and correct the mechanism before production equivalence.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-127	Verification shall inject sensor, FPGA/software, backplane, power and gate-path faults while measuring force-limit and STO response.	fault-injection matrix, STO/force timing data and traceability to risk controls
RSMC50K-128	If fault injection reveals delayed, missing or ambiguous torque removal, the system shall fail the safety release gate and correct the mechanism before production equivalence.	fault-injection matrix, STO/force timing data and traceability to risk controls
RSMC50K-129	A change to fault model, hardware safety path, FPGA logic, sensor or gate driver shall reopen fault-injection matrix, STO/force timing data and traceability to risk controls before production release.	fault-injection matrix, STO/force timing data and traceability to risk controls

Verification, margin and future-change rule

Future variants affecting fault model, hardware safety path, FPGA logic, sensor or gate driver are screened against the mechanism on this page. The release owner may retain prior evidence only where fault-injection matrix, STO/force timing data and traceability to risk controls confirms the similarity boundary.

ENVIRONMENTAL / STERILIZATION SEQUENTIAL-STRESS QUALIFICATION

Environmental qualification shall combine thermal, mechanical, EMC and approved cleaning/sterilization stresses in a sequence representative of the released lifecycle.

If post-stress electrical, calibration or safety margin is not demonstrated, the product shall block release of the affected hardware/mechanical variant until requalification is closed.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-130	Environmental qualification shall combine thermal, mechanical, EMC and approved cleaning/sterilization stresses in a sequence representative of the released lifecycle.	sequential-stress qualification, post-stress calibration and safety regression
RSMC50K-131	If post-stress electrical, calibration or safety margin is not demonstrated, the system shall block release of the affected hardware/mechanical variant until requalification is closed.	sequential-stress qualification, post-stress calibration and safety regression
RSMC50K-132	Evidence reuse after material, enclosure, PCB, connector, mechanical stack or sterilization process is permitted only when sequential-stress qualification, post-stress calibration and safety regression demonstrates that the affected assumptions remain valid.	sequential-stress qualification, post-stress calibration and safety regression

Verification, margin and future-change rule

The change trigger for this page is material, enclosure, PCB, connector, mechanical stack or sterilization process. The impact record identifies moved assumptions and the exact sequential-stress qualification, post-stress calibration and safety regression needed for closure.

SUPPLIER CHANGE, VARIANT EQUIVALENCE & FINAL RELEASE CLOSURE

Final release shall require closed precision, timing, force/STO, electrical, EMC, mechanical, manufacturing and lifecycle evidence for the exact controller variant, together with a program-approved medical compliance applicability and ownership matrix linking controller evidence to the parent surgical-system risk-management file.

Verification focuses on the point where supplier or variant evidence is incomplete or exceeds the established similarity boundary. The required containment is to withhold production-equivalence approval until targeted analysis and regression are complete.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
RSMC50K-133	Final release shall require closed precision, timing, force/STO, electrical, EMC, mechanical, manufacturing and lifecycle evidence for the exact controller variant, together with a program-approved medical compliance applicability and ownership matrix linking controller evidence to the parent surgical-system risk-management file.	release checklist, compliance applicability/ownership matrix, supplier change review and evidence index
RSMC50K-134	If supplier or variant evidence is incomplete or exceeds the established similarity boundary, production-equivalence approval shall be withheld until targeted analysis and regression are complete.	release checklist, supplier change review and evidence index
RSMC50K-135	Semiconductor, FPGA, sensor, motor, gearbox, enclosure, firmware or manufacturing site shall be treated as a controlled engineering change; approval requires release checklist, supplier change review and evidence index.	release checklist, supplier change review and evidence index

Verification, margin and future-change rule

Changes involving semiconductor, FPGA, sensor, motor, gearbox, enclosure, firmware or manufacturing site are engineering events, not editorial updates; approval depends on the targeted release checklist, supplier change review and evidence index listed here.